

Timeline Analysis & Forensic Reporting

Why Timelines Are the Heart of DFIR

A conclusion without a timeline is a gut-feeling. The timeline IS the story: event → effect, minute by minute.

Building a Super-Timeline

Merge data from many sources into one time-sorted stream:

```
# plaso/Log2Timeline – the standard
log2timeline.py case.plaso /evidence/image.dd
psort.py -o timeline case.plaso > timeline.csv
# now: every event from prefetch, registry, logs – sorted by time
```

The "Golden Hour" Analytical Window

Focus on the moments around the incident:

```
03:12:01 Failed password for root from 10.0.0.9
03:12:05 4624 An account was successfully logged on (bob, 10.0.0.9)
03:12:09 SAMService installed service 'update' (HKLM\Run)
03:12:14 powershell.exe created C:\Users\Public\p.ps1
03:12:20 Outbound HTTPS connection to 185.45.x.x:443 from powershell.exe
```

Four lines = full narrative. That's the payoff of all the artifact work.

Rabbit Holes & The Analyst's Map

Timeline gaps = missing log coverage (that's a finding itself!)

Clock skew – VM/offline hosts have wrong times; NORMALIZE before judging

Red herrings – the attacker deletes logs; the absence is evidence too

Use "around the sighting" (10 min before/after) to contextualize events

The Forensic Report (The Deliverable)

A report the C-Suite, a lawyer, AND a technician can all use:

1. EXECUTIVE SUMMARY (1 paragraph: what happened, impact, current state)
2. SCOPE & METHOD (what was examined, how, tools+hashes)
3. TIMELINE (the story in table format)
4. FINDINGS per evidence item (artifact → meaning → confidence)
5. CONCLUSION (verdict: compromised / not, at what confidence)
6. RECOMMENDATIONS (hardening, detection, staffing)
7. APPENDIX (hashes, tool versions, chain of custody)

Confidence Language (Forensics Honesty)

Confirmed – artifact directly proves it

Suspected – evidence points, but ambiguity

Unconfirmed – the story needs data we couldn't get

Forensic integrity: if you don't know, you don't guess. A report that overclaims is how the case (and your reputation) dies.

[← Windows & Linux Artifact For...](#)

Module complete ✓